AN IMMERSION OF A SYNTOMIC SCHEME INTO A SMOOTH SCHEME OVER THE BASE IS A REGULAR IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-immersion-lci-into-smooth-regular-immersion is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 5672-5684 and proof span 5686-5717 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement contains an xymatrix commutative triangle; it is copied verbatim and the retained chapter preamble loads Xy-pic.

Editorial note. 10 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A syntomic morphism is flat and locally of finite presentation, and a smooth morphism is flat and locally of finite presentation, cited from the morphisms chapter. Cited at proof lines 5686-5692. Supplies the hypotheses of the fibrewise criterion for a relative quasi-regular immersion.
- The fibrewise criterion producing a relative quasi-regular immersion from a closed subscheme whose fibres are quasi-regular, cited from the same chapter. Cited at proof lines 5692-5694. Reduces the claim to the statement that each fibre is a quasi-regular closed subscheme.
- Over a locally Noetherian scheme regularity of an immersion can be checked at points, cited from the same chapter, and a finite type scheme over a field is Noetherian. Cited at proof lines 5694-5699. Allows the fibre statement to be checked at individual points.
- The local ring of a point of a fibre of a syntomic morphism is a local complete intersection over the residue field of the base point, the local ring of a point of a smooth scheme over a field is regular, and a quotient of a regular local ring by an ideal defining a local complete intersection is cut out by a regular sequence. Cited at proof lines 5701-5711. Provides the pointwise algebra input that closes the argument.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00TT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1087–1087. Same-chapter reference used in the retained author text at source line(s) 5629, 5711. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00SF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1037–1037. Same-chapter reference used in the retained author text at source line(s) 5713. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 063I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7832–7832. Same-chapter reference used in the retained author text at source line(s) 4939, 4970, 5169, 5234, 5705. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 063W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7845–7845. Same-chapter reference used in the retained author text at source line(s) 5619, 5699. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01T6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2363–2363. Same-chapter reference used in the retained author text at source line(s) 1147, 5702. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01UG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2409–2409. Same-chapter reference used in the retained author text at source line(s) 5707. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

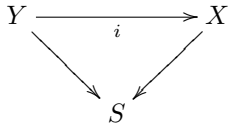
Source reference. Original lemma 01VF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2444–2444. Same-chapter reference used in the retained author text at source line(s) 5616, 5696. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01VE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2443–2443. Same-chapter reference used in the retained author text at source line(s) 5617, 5697. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01UL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2414–2414. Same-chapter reference used in the retained author text at source line(s) 5693. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01UK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2413–2413. Same-chapter reference used in the retained author text at source line(s) 5692. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let*



be a commutative diagram of morphisms of schemes. Assume that $Y \to S$ is syntomic, $X \to S$ smooth, and i an immersion. Then i is a regular immersion.

Proof. After replacing X by an open neighbourhood of $i(Y)$ we may assume that i is a closed immersion. Let $T = i(Y)$ be the corresponding closed subscheme of X. Since $T \cong Y$ the morphism $T \to S$ is flat and of finite presentation (Morphisms, Lemmas 01UK and 01UL). Also a smooth morphism is flat and locally of finite presentation (Morphisms, Lemmas 01VF and 01VE). Thus, according to Lemma 063W, it suffices to show that $T_s \subset X_s$ is a quasi-regular closed subscheme. As X_s is locally of finite type over a field, it is Noetherian (Morphisms, Lemma 01T6). Thus we can check that $T_s \subset X_s$ is a quasi-regular immersion at points, see Lemma 063I. Take $t \in T_s$. By Morphisms, Lemma 01UG the local ring $\mathcal{O}_{T_s,t}$ is a local complete intersection over $\kappa(s)$. The local ring $\mathcal{O}_{X_s,t}$ is regular, see Algebra, Lemma 00TT. By Algebra, Lemma 00SF we see that the kernel of the surjection $\mathcal{O}_{X_s,t} \to \mathcal{O}_{T_s,t}$ is generated by a regular sequence, which is what we had to show. □